

Exercise 1:

Read assembly language materials

Exercise 2:

After jumping back from 0xffff0 the BIOS then does a few things first it resets the AX register and sets the ESP register. After that the BIOS then disables interrupts and resets the direction flag. After that the BIOS starts communicating with IO devices however im not certain what it is doing. Im assuming it is detecting / configuring present hardware.

Exercise 3:

First I found bootmain() in the boot.asm file, here is bootmain in assembly

```

00007d0a <bootmain>:
{
  7d0a: 55                push    %ebp
  7d0b: 89 e5            mov     %esp,%ebp
  7d0d: 56              push    %esi
  7d0e: 53              push    %ebx
  readseg((uint32_t) ELFHDR, SECTSIZE*8, 0);
  7d0f: 6a 00            push    $0x0
  7d11: 68 00 10 00 00    push    $0x1000
  7d16: 68 00 00 01 00    push    $0x10000
  7d1b: e8 b1 ff ff ff    call    7cd1 <readseg>
  if (ELFHDR->e_magic != ELF_MAGIC)
  7d20: 83 c4 0c          add     $0xc,%esp
  7d23: 81 3d 00 00 01 00 7f  cml     $0x464c457f,0x10000
  7d2a: 45 4c 46          jne     7d67 <bootmain+0x5d>
  ph = (struct Proghdr *) ((uint8_t *) ELFHDR + ELFHDR->e_phoff);
  7d2f: a1 1c 00 01 00    mov     0x1001c,%eax
  7d34: 8d 98 00 00 01 00    lea     0x10000(%eax),%ebx
  eph = ph + ELFHDR->e_phnum;
  7d3a: 0f b7 05 2c 00 01 00  movzwl  0x1002c,%eax
  7d41: c1 e0 05          shl     $0x5,%eax
  7d44: 8d 34 03          lea     (%ebx,%eax,1),%esi
  for (; ph < eph; ph++)
  7d47: 39 f3            cmp     %esi,%ebx
  7d49: 73 16            jae     7d61 <bootmain+0x57>
    readseg(ph->p_pa, ph->p_memsz, ph->p_offset);
  7d4b: ff 73 04          pushl   0x4(%ebx)
  for (; ph < eph; ph++)
  7d4e: 83 c3 20          add     $0x20,%ebx
    readseg(ph->p_pa, ph->p_memsz, ph->p_offset);
  7d51: ff 73 f4          pushl   -0xc(%ebx)
  7d54: ff 73 ec          pushl   -0x14(%ebx)
  7d57: e8 75 ff ff ff    call    7cd1 <readseg>
  for (; ph < eph; ph++)
  7d5c: 83 c4 0c          add     $0xc,%esp
  7d5f: eb e6            jmp     7d47 <bootmain+0x3d>
  ((void (*)(void)) (ELFHDR->e_entry))();
  7d61: ff 15 18 00 01 00    call    *0x10018
}

```

Then I repeated the process to find the readsect() section

The red box represents a line of c code that can be found in readsect() in main.c. The yellow box immediately after a red box represents the assembly that corresponds to the c code.

```

155 00007c7c <readsect>:
156
157 void
158 readsect(void *dst, uint32_t offset)
159 {
160     7c7c: 55                push    %ebp
161     7c7d: 89 e5            mov     %esp,%ebp
162     7c7f: 57                push    %edi
163     7c80: 53                push    %ebx
164     7c81: 8b 5d 0c          mov     0xc(%ebp),%ebx
165     // wait for disk to be ready
166     waitdisk()
167     7c84: e8 e1 ff ff ff    call    7c6a <waitdisk>
168 }
169
170 static inline void
171 outb(int port, uint8_t data)
172 {
173     asm volatile("outb %0,%w1" : : "a" (data), "d" (port))
174     7c89: ba +2 01 00 00    mov     $0x1+2,%edx
175     7c8e: b0 01            mov     $0x1,%al
176     7c90: ee                out     %al, (%dx)
177     7c91: 0f b6 c3          movzbl  %bl,%eax
178     7c94: b2 f3            mov     $0xf3,%dl
179     7c96: ee                out     %al, (%dx)
180     7c97: 0f b6 c7          movzbl  %bh,%eax
181     7c9a: b2 f4            mov     $0xf4,%dl
182     7c9c: ee                out     %al, (%dx)
183

```

```

183
184 outb(0x1F2, 1); // count = 1
185 outb(0x1F3, offset);
186 outb(0x1F4, offset >> 8);
187 outb(0x1F5, offset >> 16);
188 7c9d: 89 d8      mov     %ebx,%eax
189 7c9f: b2 f5      mov     $0xf5,%dl
190 7ca1: c1 e8 10    shr     $0x10,%eax
191 7ca4: 0f b6 c0    movzbl %al,%eax
192 7ca7: ee         out     %al,(%dx)
193 outb(0x1F6, (offset >> 24) | 0xE0);
194 7ca8: c1 eb 18    shr     $0x18,%ebx
195 7cab: b2 f6      mov     $0xf6,%dl
196 7cad: 88 d8      mov     %bl,%al
197 7caf: 83 c8 e0    or      $0xffffffe0,%eax
198 7cb2: ee         out     %al,(%dx)
199 7cb3: b0 20      mov     $0x20,%al
200 7cb5: b2 f7      mov     $0xf7,%dl
201 7cb7: ee         out     %al,(%dx)
202 outb(0x1F7, 0x20); // cmd 0x20 - read sectors
203
204 // wait for disk to be ready
205 waitdisk()
206 7cb8: e8 ad ff ff ff call    7c6a <waitdisk>
207 asm volatile("cld\n\trepne\n\tinsl")
208 7cbd: 80 7d 08    mov     0x8(%ebp),%edi
209 7cc0: b9 80 00 00 00 mov     $0x80,%ecx
210 7cc5: ba f0 01 00 00 mov     $0x1f0,%edx
211 7cca: fc         cld
212 7ccb: f2 6d      repnz  insl (%dx),%es:(%edi)
213
214 // read a sector
215 insl(0x1F0, dst, SECTSIZE/4);
216 }
217 7ccd: 5b         pop     %ebx
218 7cce: 5f         pop     %edi
219 7ccf: 5d         pop     %ebp
220 7cd0: c3         ret
221

```

Exercise 4:

Reviewed how pointers work.

Exercise 5:

The first instruction that will break is the **lgdt** instruction because it loads data from an absolute address.

In my case the program went to an infinite loop because a fault was triggered.

Exercise 6

When bios enter boot loader

(gdb) x/8x 0x00100000

```

0x100000:  0x00000000  0x00000000  0x00000000  0x00000000  0x00000000
0x100010:  0x00000000  0x00000000  0x00000000  0x00000000  0x00000000

```

When boot loader enter kernel

(gdb) x/10x 0x00100000

0x100000:	0x1badb002	0x00000000	0xe4524ffe	0x7205c766
0x100010:	0x34000004	0x2000b812	0x220f0011	0xc0200fd8
0x100020:	0x0100010d	0xc0220f80		

The set of data from the first breakpoint is all 0s because the kernel hasn't been loaded. The data at the second breakpoint contains information about the kernel

Exercise 7

The first instruction after the new mapping that would fail is:

f010002d: ff e0 jmp *%eax

This is because without paging it will jump to an invalid physical address or an address of all zeros.

Exercise 8

Code inserted

```
// (unsigned) octal
case 'o':
    // LAB 1: Replace this with your code.
    num = getuint(&ap, lflag);
    base = 8;
    goto number;
```

```
qemu-system-i386 -nographic -drive file=obj/kern/kernel
444544 decimal is 1544200 octal!
```

Exercise 9

The stack is initialized here in entry.S

```
# Set the stack pointer
movl    $(bootstacktop), %esp
```

The address is shown in the .data section. The stack grows downward so bootstacktop is where the stack pointer will start at and grow down to the lower address of the .data section.

Bootstacktop:

	CONTENTS	ALLOC	LOAD	READONLY	DATA
4 .data	0000a300	f0108000	00108000	00009000	2**12

Lower end of .data

```
movl    $(bootstacktop), %esp
f0100034: bc 00 00 11 f0          mov     $0xf0110000, %esp
```

So the stack will be allocated from 0xf0108000 to 0xf0110000. \$esp will start at 0xf0110000

Exercise 10

From looking at kernel.asm I put a break point at 0xf0100040

```
@arku@fgh:
-- Output/Registers
The current architecture is assumed to be i386
-> 0xf0100040: test_backtrace: push    ebp
13: i
-- Registers
rax 0x00000000  ebx 0x00000000  ecx 0x00000000  edx 0x00000000  esi 0x00000000  edi 0x00000000  eip 0xf0100040  [flag: 0x0]
-- Assembly
0xf0100040: test_backtrace+0: push    ebp
0xf0100041: test_backtrace+1: mov     ebp, %esp
0xf0100042: test_backtrace+2: push    esi
0xf0100043: test_backtrace+3: mov     esi, %eax
0xf0100044: test_backtrace+4: mov     edi, %eax
0xf0100045: test_backtrace+5: mov     ebx, %eax
-- Source
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Exercise 11

Implemented backtrace function.

Exercise 12

Modified backtrace function to display for each eip the function name, source file, and line number.